public static int[] Merge_Two_Array(int[] arr1, int[] arr2) { int n = arr1.length;
int m = arr2.length;
int[] ans = new int[n + m];
int i = 0, j = 0, k = 0;
while (i < n && j < m) {
if (arr1[i] < arr2[j]) {
ans[k++] = arr1[i++];

}
else {
ans[k++] = arr2[j++];
}
}
}

int[] arr = { 5, 7, 2, 1, 8, 3, 4 };

public static int[] Sort(int[] arr, int si, int ei) {
if (si == ei) {
int[] bs = new int[1];
bs[0] = arr[si];
return bs;
}
int mid = (si + ei) / 2;
int[] a = Sort(arr, si, mid);
int[] b = Sort(arr, mid + 1, ei);
return Merge_Two_Array(a, b);

}

public static int Partition(int[] arr, int si, int ei) {
int item = arr[ei];
int idx = si;
for (int i = si; i < ei; i++) {
if (arr[i] < item) {
int temp = arr[idx];
arr[idx] = arr[i];
arr[i] = temp;
idx++;
}
}
int temp = arr[idx];
arr[idx] = arr[ei];
arr[ei] = temp;
return idx;

}

public static int fact(int n) {
if (n == 0) {

return 1;

}

int fn = fact(n - 1);// SP

return fn * n;
}

public static int fib(int n) {
if (n == 0 || n == 1) {
return n;

}

int f1 = fib(n - 1);
int f2 = fib(n - 2);
return f1 + f2;

}

public static void Sort(int[] arr, int si, int ei) {
if (si >= ei) {
return;
}
int idx = Partition(arr, si, ei);
Sort(arr, si, idx - 1);
Sort(arr, idx + 1, ei);
}

public static int Partition(int[] arr, int si, int ei) {
int item = arr[ei];
int idx = si;
for (int i = si; i < ei; i++) {
if (arr[i] < item) {
int temp = arr[idx];
arr[idx] = arr[i];
arr[i] = temp;
idx++;
}
}
int temp = arr[idx];
arr[idx] = arr[ei];
arr[ei] = temp;
return idx;

}

public static int pow(int a, int n) {
if(n==0) {
return 1;
}
int ans=pow(a,n/2);
ans= ans*ans;
if(n%2!=0) {
ans=ans*a;
}
return ans;

}

public static int fact(int n) {
if (n == 0) {

return 1;

}

int fn = fact(n - 1);// SP
return fn * n;
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public static int pow(int a, int n) {
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public static int fib(int n) {

if (n == 0 || n == 1) {

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int f1 = fib(n - 1);

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return f1 + f2;

}

public static int fib(int n) {

if (n == 0 || n == 1) {

return n;

}

int f1 = fib(n - 1);

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return f1 + f2;

}

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